AI Copilot Usage Log

Project: OzZoo GUI Simulation

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Curated record of key prompts used for design, implementation, debugging, optimisation, and validation. This is intentionally selective rather than a full transcript.

1. Visitor reviews affect game state and reputation

Key Prompt

> Now reviews from our visitors are not affecting the game state. Instructions: when each day advances, add the reviews to the pop-up dialog. Visitors are more likely to leave positive reviews for happy animals, and bad reviews for neglected animals. Add a rating like '4 out of 5 stars' and add 'Reputation' next to 'Welfare' on the Dashboard.

AI Output Snippet

```
visitor_count_for_reviews = len(self._visitors)
if self._day > 1 and visitor_count_for_reviews > 0:
    daily_reviews = generate_reviews(
        self._animals,
        self._day,
        visitor_count=visitor_count_for_reviews
    )
else:
    daily_reviews = []
self._reviews.extend(daily_reviews)
report['reviews'] = daily_reviews
self._reputation = calculate_reputation_from_reviews(self._reviews)
```

Your Analysis

This prompt directly targeted a simulation gap. I used it to connect daily visitor feedback into persistent state so reviews influence future gameplay through reputation.

Source: ozzoo/models/zoo.py

2. Mission progression fix and mission lifecycle handling

Key Prompt

> okay now lets fix the mission progression, there is no progress recorded when the mission requirements are satified when the day advances. I would also like to give the player a new mission when they have failed or completed their current one.

AI Output Snippet

```
def update_mission_progress(self, include_types=None) -> None:
    if not self._active_missions:
        return
    if include_types is None or MissionType.KEEP_HAPPY in include_types:
        self._update_global_happy_streaks()
    for mission in self._active_missions:
        if include_types is not None and mission.mission_type not in include_types:
            continue
        self._update_single_mission_progress(mission)

def _resolve_active_missions(self) -> tuple[list[str], list[str]]:
    for mission in list(self._active_missions):
        if mission.is_complete:
            self.complete_mission(mission)
        elif mission.is_failed:
            self._active_missions.remove(mission)
```

Your Analysis

I used this to repair progression timing at day boundaries and ensure missions exit cleanly on completion/failure, preventing stale mission states.

Source: ozzoo/models/zoo.py

3. Stacking missions every 3 in-game days

Key Prompt

Feature: New mission every 3 days (stacking). Missions stack and multiple can be active simultaneously.

AI Output Snippet

```
def add_mission(self, mission: Mission) -> bool:
    if any(existing.mission_id == mission.mission_id for existing in self._active_missions):
        return False
    self._active_missions.append(mission)
    return True

def _add_periodic_mission_if_due(self) -> Optional[Mission]:
    if self._day % 3 != 0:
        return None
    active_types = {
        mission.mission_type
        for mission in self._active_missions
        if not mission.is_complete and not mission.is_failed
    }
    mission = generate_random_mission(..., exclude_types=active_types)
    if self.add_mission(mission):
        return mission
    return None
```

Your Analysis

This prompt was used to move from a single-mission model to concurrent missions. The design avoids duplicate active mission types while preserving unfinished tasks.

Source: ozzoo/models/zoo.py

4. Visitor attraction tuning with animal diversity

Key Prompt

Lets add, animal diversity. and I think we should remove the zoo age/progression.

AI Output Snippet

```
def generate_daily_visitors(self) -> int:
    base_visitors = 20
    welfare_bonus = int(self.average_animal_welfare / 10)
    diversity_bonus = min(18, max(0, self.species_diversity - 1) * 2)
    sick_penalty = self.sick_animal_count * 2
    num_visitors = max(5, base_visitors + welfare_bonus + diversity_bonus - sick_penalty)
```

Your Analysis

I used this to simplify the attraction model and make it strategy-driven: players are rewarded for species variety and welfare rather than passive zoo age.

Source: ozzoo/models/zoo.py

5. Mouse-wheel scrolling for ShopFrame

Key Prompt

The Zoo ShopFrame needs to be mouse-wheel scrollable.

AI Output Snippet

```
def _enable_mousewheel_scrolling(self) -> None:
    if not self._scroll_canvases:
        return
    def _get_target_canvas() -> tk.Canvas | None:
        hovered = self.winfo_containing(self.winfo_pointerx(), self.winfo_pointery())
        ...
    def _on_mousewheel(event: tk.Event) -> str:
        canvas = _get_target_canvas()
        if canvas is None or event.delta == 0:
            return 'break'
        step = -1 if event.delta > 0 else 1
        canvas.yview_scroll(step, 'units')
```

```
        return 'break'
    self.bind_all('<MouseWheel>', _on_mousewheel)
```

Your Analysis

This prompt focused on UX polish. The solution routes wheel events to the currently hovered canvas so scrolling works reliably across tabs.

Source: ozzoo/ui/frames/shop.py

6. Predation framework for snake predator/prey pairs

Key Prompt

> Add new reptile predator/prey behaviour and follow existing predation risk infrastructure throughout.

AI Output Snippet

```
REPTILE_PREDATOR_PREY_BY_PREDATOR = {
    'CarpetPython': frozenset({'RingtailPossum', 'FrilledLizard'}),
    'EasternBrownSnake': frozenset({'RingtailPossum', 'BlueTonguedSkink'}),
}
REPTILE_PREDATOR_SPECIES = frozenset(REPTILE_PREDATOR_PREY_BY_PREDATOR.keys())
REPTILE_PREDATOR_PREY_SPECIES = frozenset(
    prey_species
    for prey_set in REPTILE_PREDATOR_PREY_BY_PREDATOR.values()
    for prey_species in prey_set
)
```

Your Analysis

I used this prompt to keep predator/prey logic explicit and extensible. A pair-mapping structure allows future predator additions without rewriting the core risk pipeline.

Source: ozzoo/models/zoo.py

7. Tasmanian Devil mixed-species outback risk

Key Prompt

> lets add predation risk when a player adds a Tasmanian Devil to the Outback Plains containing any of the other species.

AI Output Snippet

```
def _get_outback_devil_prey_animals(self, enclosure: Enclosure) -> list[Animal]:
    if enclosure.get_habitat_type() != 'outback_savanna':
        return []
    has_devil = any(a.is_alive and a.species == 'TasmanianDevil' for a in enclosure.animals)
    if not has_devil:
        return []
    return [a for a in enclosure.animals if a.is_alive and a.species != 'TasmanianDevil']
```

Your Analysis

This prompt introduced a habitat-specific danger rule. The implementation detects mixed-species exposure and feeds that into the shared daily risk engine.

Source: ozzoo/models/zoo.py

8. Goanna compatibility in Reptile House

Key Prompt

> Ensure the Goanna can also be added to the Reptile House.

AI Output Snippet

```
class EucalyptusGrove(Enclosure):
    def get_compatible_species(self) -> List[str]:
        return ['Koala', 'RingtailPossum', 'Goanna', 'CarpetPython', 'EasternBrownSnake']

class ReptileHouse(Enclosure):
    def get_compatible_species(self) -> List[str]:
        return ['FrilledLizard', 'BlueTonguedSkink', 'Goanna', 'CarpetPython', 'EasternBrownSnake']
```

Your Analysis

I used this prompt to support multi-habitat species placement. Compatibility lists were updated so Goanna works in both intended enclosure types.

Source: ozzoo/models/enclosure.py

9. Keeper safety mishap for Eastern Brown Snake neglect

Key Prompt

Add keeper safety mishap behaviour for Eastern Brown Snake under neglected maintenance.

AI Output Snippet

```
def _process_eastern_brown_keeper_safety_events(self, report: dict[str, Any]) -> list[str]:
    for enclosure in self._enclosures.values():
        eastern_browns = [a for a in enclosure.animals if a.is_alive and a.species == 'EasternBrownSnake']
        if not eastern_browns:
            continue
        neglected_maintenance = (
            enclosure.cleanliness <= 45
            or (enclosure.is_dirty and enclosure.barrier_level == 0)
        )
        ...
        incident_cost = round(self.EASTERN_BROWN_KEEPER_MISHAP_COST * (1 + cleanliness_severity), 2)
        self._budget -= incident_cost
        self._total_expenses += incident_cost
        report['expenses'] += incident_cost
```

Your Analysis

This prompt added systemic consequences for poor enclosure maintenance, linking husbandry quality to staff safety, finances, and visitor sentiment.

Source: ozzoo/models/zoo.py

10. Runtime verification prompt

Key Prompt

can retry Python runtime checks

AI Output Snippet

```
python -m compileall -q ozzoo
python -c "from ozzoo.models.zoo import create_default_zoo; z=create_default_zoo(); report=z.advance_day();
Output: advanced_day=1 events=0 deaths=0
```

Your Analysis

I used this prompt to confirm the integrated systems still ran end-to-end after species and predation expansions.

Source: Runtime command log